

Module 3 and 4 (Schema and attributes)

Module 3: Supplier & Purchase Management

1. Suppliers

Stores supplier/vendor information.

Column	Data Type	Constraints
supplier_id	INT	PK, AUTO_INCREMENT
supplier_name	VARCHAR(100)	NOT NULL
contact_person	VARCHAR(100)	NULL
phone	VARCHAR(20)	NOT NULL
email	VARCHAR(100)	NULL
address	TEXT	NULL
gst_number	VARCHAR(20)	NULL
created_at	DATETIME	DEFAULT CURRENT_TIMESTAMP

2. medicines

Basic/master information about medicines.

Column	Data Type	Constraints
medicine_id	INT	PK, AUTO_INCREMENT
medicine_name	VARCHAR(150)	NOT NULL
generic_name	VARCHAR(150)	NULL
category	VARCHAR(100)	NULL
manufacturer	VARCHAR(150)	NULL
unit	VARCHAR(30)	NOT NULL

reorder_level	INT	DEFAULT 0
created_at	DATETIME	DEFAULT CURRENT_TIMESTAMP

3. purchase_orders

Stores the main purchase order.

Column	Data Type	Constraints
purchase_id	INT	PK, AUTO_INCREMENT
supplier_id	INT	FK → suppliers.supplier_id
purchase_date	DATE	NOT NULL
status	VARCHAR(20)	DEFAULT 'PENDING'
total_amount	DECIMAL(10,2)	DEFAULT 0
created_at	DATETIME	DEFAULT CURRENT_TIMESTAMP

Status examples: PENDING, RECEIVED, CANCELLED.

4. purchase_items

Stores individual medicines within a purchase order.

Column	Data Type	Constraints
purchase_item_id	INT	PK, AUTO_INCREMENT
purchase_id	INT	FK → purchase_orders.purchase_id
medicine_id	INT	FK → medicines.medicine_id
quantity	INT	NOT NULL
purchase_price	DECIMAL(10,2)	NOT NULL
subtotal	DECIMAL(10,2)	NOT NULL

This allows one purchase order to contain multiple medicines.

5. batches

Stores batch-wise medicine information.

Column	Data Type	Constraints
batch_id	INT	PK, AUTO_INCREMENT
medicine_id	INT	FK → medicines.medicine_id
purchase_item_id	INT	FK → purchase_items.purchase_item_id
batch_number	VARCHAR(50)	NOT NULL
manufacturing_date	DATE	NULL
expiry_date	DATE	NOT NULL
quantity	INT	NOT NULL
purchase_price	DECIMAL(10,2)	NOT NULL
selling_price	DECIMAL(10,2)	NOT NULL

This table is important because the same medicine can have multiple batches with different expiry dates and prices.

6. stock

Stores the current inventory quantity.

Column	Data Type	Constraints
stock_id	INT	PK, AUTO_INCREMENT
medicine_id	INT	FK → medicines.medicine_id, UNIQUE
quantity	INT	DEFAULT 0
updated_at	DATETIME	DEFAULT CURRENT_TIMESTAMP

Module 4: Billing / Point-of-Sale

7. billing_queue

Handles the FIFO customer queue.

Column	Data Type	Constraints
queue_id	INT	PK, AUTO_INCREMENT
token_number	INT	NOT NULL
customer_name	VARCHAR(100)	NULL
joined_at	DATETIME	DEFAULT CURRENT_TIMESTAMP
status	VARCHAR(20)	DEFAULT 'WAITING'
served_at	DATETIME	NULL

Status: WAITING, SERVING, COMPLETED, CANCELLED.

FIFO can be implemented using joined_at or an increasing queue_id.

8. sales

Stores the main invoice/bill.

Column	Data Type	Constraints
sale_id	INT	PK, AUTO_INCREMENT
queue_id	INT	FK → billing_queue.queue_id, NULL
invoice_number	VARCHAR(50)	UNIQUE, NOT NULL
sale_date	DATETIME	DEFAULT CURRENT_TIMESTAMP
subtotal	DECIMAL(10,2)	DEFAULT 0
discount	DECIMAL(10,2)	DEFAULT 0
tax	DECIMAL(10,2)	DEFAULT 0

total_amount	DECIMAL(10,2)	NOT NULL
payment_method	VARCHAR(20)	NOT NULL
status	VARCHAR(20)	DEFAULT 'COMPLETED'

Payment methods: CASH, CARD, UPI, etc.

9. sale_items

Stores medicines included in each bill.

Column	Data Type	Constraints
sale_item_id	INT	PK, AUTO_INCREMENT
sale_id	INT	FK → sales.sale_id
medicine_id	INT	FK → medicines.medicine_id
batch_id	INT	FK → batches.batch_id
quantity	INT	NOT NULL
selling_price	DECIMAL(10,2)	NOT NULL
discount	DECIMAL(10,2)	DEFAULT 0
subtotal	DECIMAL(10,2)	NOT NULL